



SCHOOL OF COMPUTER SCIENCES

UNIVERSITI SAINS MALAYSIA

CPC351: Principles of Data Analytics

Semester I, Academic Session: 2024/2025

Project

Group 19

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Date of Submission:

19th of January 2025

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1.0 Introduction

Weather pattern analysis and prediction have become increasingly important in various sectors, from agriculture to urban planning. The ability to understand and classify weather patterns not only helps in daily operations but also contributes to decision-making processes. Hence, this project focuses on analysing weather data using machine learning techniques to identify distinct weather patterns and develop a classification model for future predictions.

1.1 Problem Statement

Weather stations collect vast amounts of meteorological data, such as temperature, humidity, and barometric pressure. However, raw data alone provides limited insights into weather patterns, which are inherently complex and challenging to predict. The challenge lies in identifying distinct weather patterns, developing a reliable classification system, and creating a model that accurately predicts these patterns from meteorological measurements.

1.2 Objectives

The main objectives of this project are:

1. Develop an automated weather pattern classification system using unsupervised learning techniques.
2. Create a predictive model for weather pattern identification using supervised learning.
3. Evaluate the effectiveness of combining clustering and classification approaches for weather analysis.

1.3 Hypothesis

The primary hypotheses for this project are:

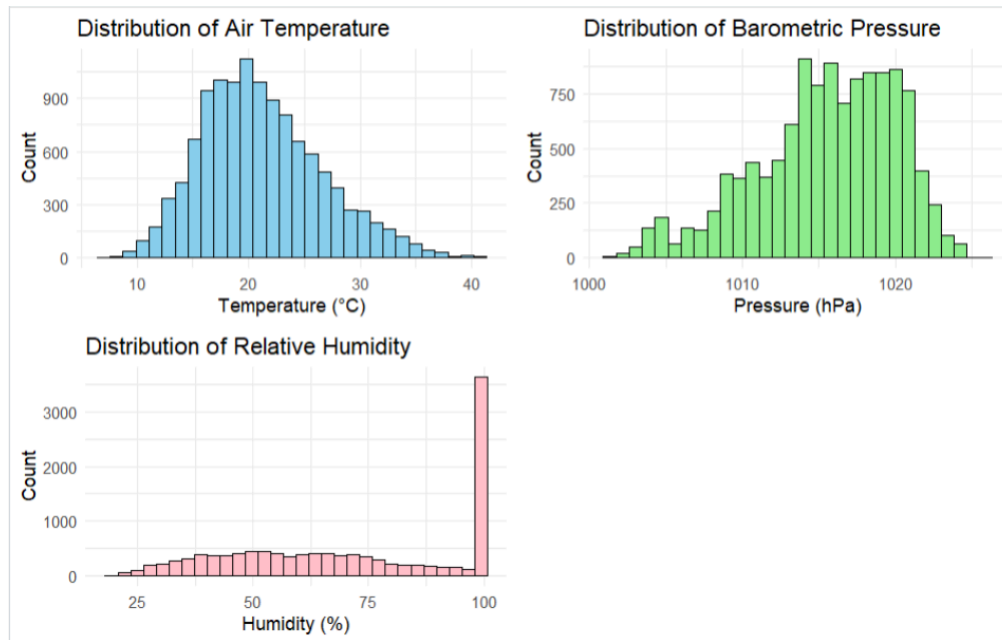
1. K-means clustering can identify distinct and interpretable weather patterns from meteorological data.
2. Naive Bayes classification will achieve high accuracy (>80%) in predicting weather patterns based on these clusters.
3. Meteorological variables (temperature, humidity, and pressure) contribute differently to determining weather patterns.

2.0 Data Preparation

The original weather dataset used in this analysis contains 12,945 records from micro-climate sensors in the City of Canning, with seven variables. Understanding and properly preparing this data is crucial for subsequent analysis and modeling. By ensuring data quality through thorough checks for missing values, duplicate entries, and outliers, we can enhance the reliability of our findings and support more accurate predictive analytics.

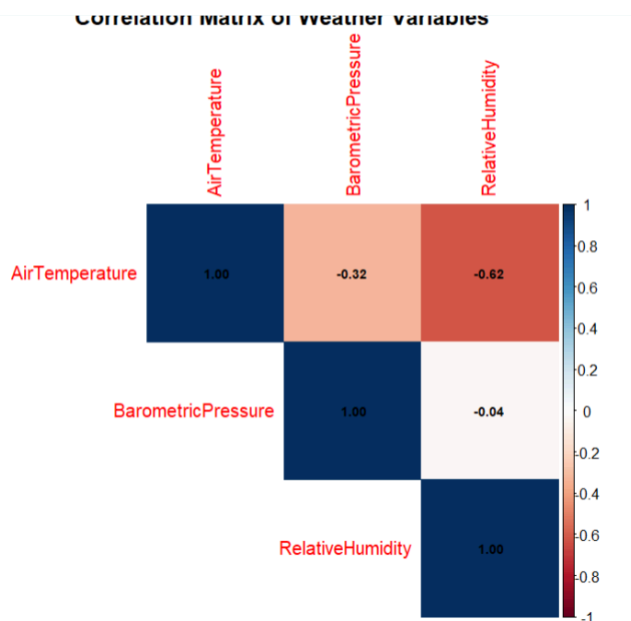
2.1 Exploratory Data Analysis (EDA)

From initial examination of the dataset, it revealed these three primary meteorological variables: Air Temperature (°C), Barometric Pressure (hPa), and Relative Humidity (%). The dataset spans multiple time periods, with measurements taken at various intervals throughout the day.



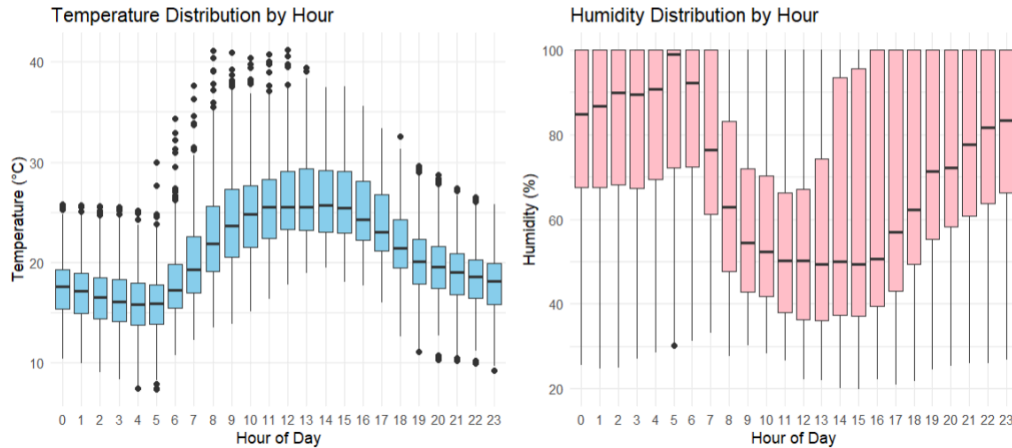
[Figure 2.1: Distribution plots of main weather variables]

The distributions reveal that air temperature has a fairly regular distribution, although relative humidity has significantly skewed patterns.



The correlation analysis shows a substantial negative association between temperature and humidity (-0.78), implying that higher temperatures are usually associated with lower humidity levels. Barometric pressure exhibited weak associations with temperature and humidity.

[Figure 2.2: Correlation matrix heatmap]



[Figure 2.3: Box plots of temperature and humidity by hour]

The box plots reveal distinct diurnal trends, with temperature peaking in the afternoon and humidity displaying an inverse relationship.

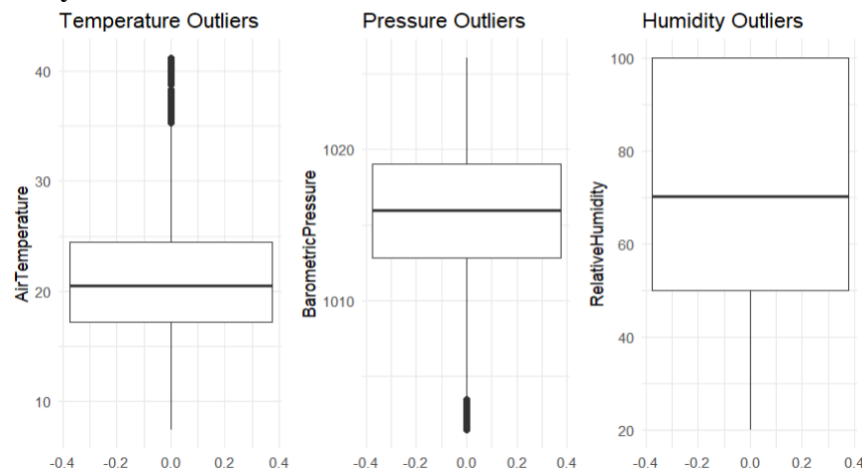
2.2 Data Pre-processing and Cleaning

2.2.1 Data Quality Assessment

In the early steps of processing data, we need to identify the data quality that can cause several issues:

1. Confirmed data completeness: No missing values were found in the dataset
2. Removed duplicate entries: 1157 duplicate rows were identified and removed
3. Validated the timestamp formats and converted to appropriate datetime objects

2.2.2 Outlier Analysis



[Figure 2.4: Box plots showing outlier detection]

The outlier analysis identified:

- 114 temperature outliers
- 79 pressure outliers
- 0 humidity outliers

2.2.3 Feature Engineering

Several derived features were created to enhance the analysis:

1. Created temporal features from timestamps:
 - Hour of the day
 - Month
 - Day of week
2. Added derived categorical variables:
 - Time of day categories (Morning, Afternoon, Evening, Night)
 - Season (Summer, Autumn, Winter, Spring)

2.2.4 Data Validation

- Confirmed all temperature readings within reasonable ranges
- Ensured that humidity measurements were consistently within the standard range of 0% to 100%, reflecting valid sensor readings.
- Confirmed barometric pressure readings aligned with typical atmospheric pressure levels,

3.0 Model Planning and Development

Based on our objectives to understand and predict weather patterns, we implemented a two-stage modeling approach. First, we used K-means clustering to identify natural weather patterns in the data. The clustering results then served as labels for a Naive Bayes classifier, creating an integrated analysis pipeline.

3.1 Model Selection

K-means Clustering:

K-means clustering was chosen for its ability to identify natural groupings in continuous meteorological data, offering interpretable cluster centers that represent distinct weather patterns. It is computationally efficient, making it suitable for large datasets and identifying patterns in numerical weather parameters.

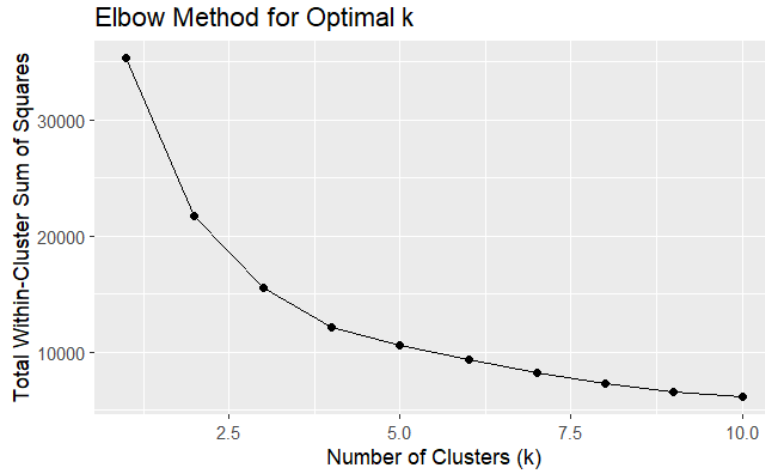
Naive Bayes Classification:

Naive Bayes classification was selected for its effectiveness with numerical and categorical features and its ability to handle the probabilistic nature of weather pattern prediction. It is computationally efficient, scalable, and provides probabilistic predictions useful for classifying weather patterns.

3.2 Model Implementation

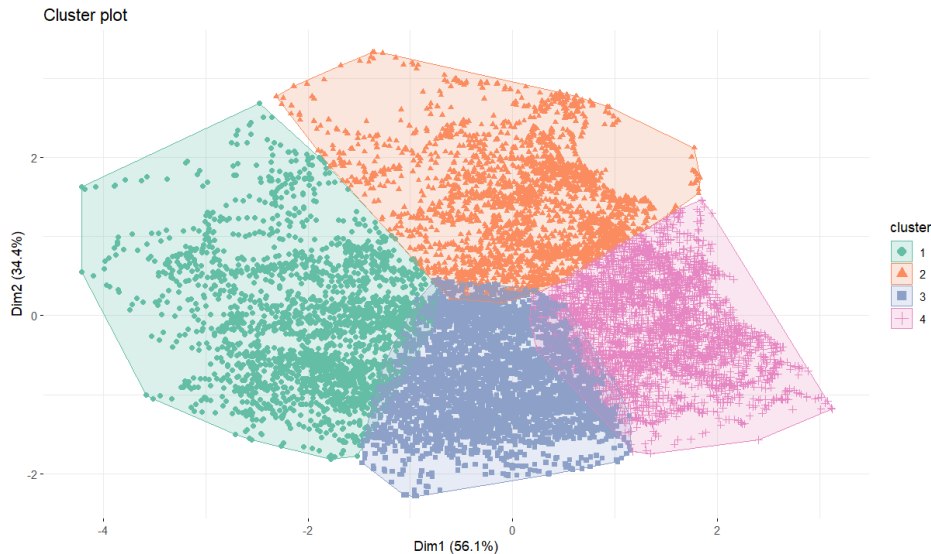
3.2.1 K-means Clustering

To identify natural groupings in the weather data, the features Air Temperature, Barometric Pressure, and Relative Humidity were selected and scaled. The optimal number of clusters was determined using the elbow method.



[Figure 3.1: Elbow plot for optimal k determination]

The elbow plot suggested 4 clusters as optimal, with 3 as a strong contender. The clustering itself was performed using three standardised weather variables: *Air Temperature*, *Barometric Pressure* and *Relative Humidity*. K-means clustering was applied with four centers, and the resulting clusters were characterised by their average temperature, humidity, and pressure values.



[Figure 3.2: Cluster visualisation plot]

The `fviz_cluster()` function uses Principal Component Analysis (PCA) to reduce the three dimensions (*AirTemperature*, *BarometricPressure*, *RelativeHumidity*) into two principal components to allow visualisation in a 2D plot. The visualisation shows clear separation between the identified weather patterns, validating our clustering approach.

3.2.2 Naive Bayes Classification

The Naive Bayes model was implemented using features selected through RFE, including temperature, humidity, pressure, and temporal attributes. Data preprocessing included standardisation of numeric features to meet algorithmic requirements. The dataset was split into training, validation, and test sets (60%, 20%, 20%), and classification performance was evaluated using confusion matrices and accuracy metrics.

3.3 Model Technical Details

K-means Clustering:

- Optimal clusters, k: 4 (determined through elbow method analysis)
- Distance metric: Euclidean
- Initialisation: Random initialisation (seeded to maintain replication)
- Maximum iterations: 25 different random starts
- Input features: AirTemperature, BarometricPressure, RelativeHumidity

Naive Bayes:

- Type: Gaussian Naive Bayes (Using e1071's library naiveBayes function)
- Data splitting: 60% training, 20% validation, 20% test set
- Features: Selected through RFE (Recursive Feature Elimination) and standardised
- Cross-validation: 5-fold CV during feature selection

4.0 Results and Discussion

Our analysis revealed significant insights into the weather patterns of the City of Canning through both clustering and classification approaches.

4.1 Clustering Results

The K-means clustering analysis identified four distinct weather patterns:

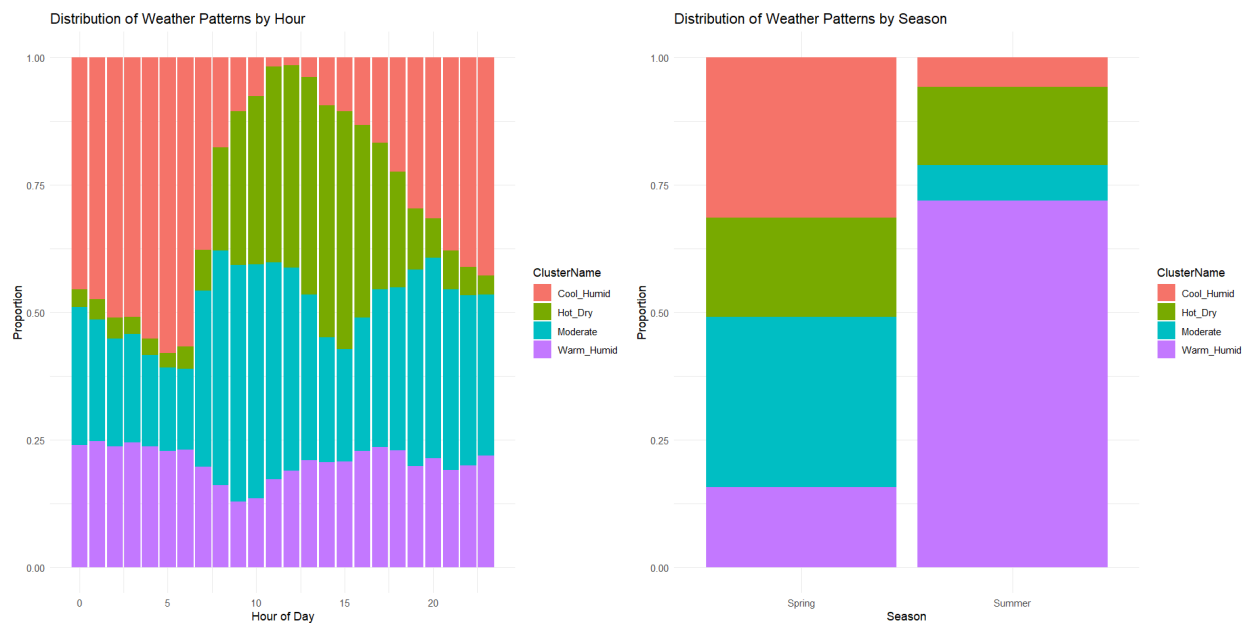
```
2. Clustering Results:
# A tibble: 4 x 5
  ClusterName avg_temp avg_pressure avg_humidity count
<fct>         <dbl>         <dbl>         <dbl> <int>
1 Cool_Humid    16.2          1018.          93.5  3438
2 Hot_Dry       28.9          1013.          40.1  2249
3 Moderate      20.9          1018.          56.3  3656
4 Warm_Humid    21.3          1010.          90.1  2445
```

[Figure 4.1: Cluster characteristics summary plot]

Each cluster displayed distinct characteristics:

1. Hot_Dry: High temperatures (28.9°C) combined with relatively low humidity (40.1%).
2. Warm_Humid: Moderate temperatures (21.3°C) and elevated humidity levels (90.1%).
3. Moderate: Balanced conditions with moderate temperatures (20.9°C), pressure (1018.0 hPa), and humidity (56.3%).
4. Cool_Humid: Low temperatures (16.2°C) and high relative humidity (93.5%).

Temporal Distribution of Weather Patterns:



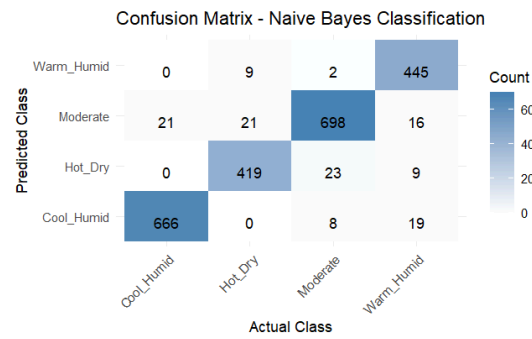
[Figure 4.2: Weather pattern distribution by hour and season]

The temporal analysis of weather patterns revealed distinct variations based on the time of day and the season. As shown in the distribution by hour, Cool_Humid patterns were predominantly observed during the early morning hours, gradually decreasing as the day progressed. In contrast, Hot_Dry patterns increased significantly in the afternoon, corresponding to rising temperatures. Moderate and Warm_Humid patterns displayed more consistent proportions throughout the day, with slight peaks during late mornings and evenings.

Seasonally, Warm_Humid conditions were more dominant during summer, contributing to over half the total observations. Cool_Humid and Moderate patterns however, contended with one another for the majority of spring.

4.2 Classification Performance

The Naive Bayes classifier demonstrated strong predictive capability:



[Figure 4.3: Confusion matrix visualisation]

Performance Metrics:

Overall Statistics				
Accuracy : 0.9457				
95% CI : (0.9357, 0.9545)				
No Information Rate : 0.3103				
P-Value [Acc > NIR] : < 2.2e-16				
Kappa : 0.9264				
McNemar's Test P-Value : NA				
Statistics by Class:				
	Class: Cool_Humid	Class: Hot_Dry	Class: Moderate	Class: Warm_Humid
Sensitivity	0.9694	0.9332	0.9549	0.9100
Specificity	0.9838	0.9832	0.9643	0.9941
Pos Pred Value	0.9610	0.9290	0.9233	0.9759
Neg Pred Value	0.9874	0.9843	0.9794	0.9768
Prevalence	0.2916	0.1906	0.3103	0.2076
Detection Rate	0.2827	0.1778	0.2963	0.1889
Detection Prevalence	0.2941	0.1914	0.3209	0.1935
Balanced Accuracy	0.9766	0.9582	0.9596	0.9521

The model exhibited the highest sensitivity for the **Cool_Humid** class (96.94%) and the highest precision for the **Warm_Humid** class (97.59%). The balanced accuracy for all classes remained consistently above 95%, underscoring the model's ability to handle imbalanced prevalence rates effectively. The high specificity (above 98%) across classes further highlights its capability to minimise false positives.

Overall, the Naive Bayes classifier proved to be highly effective in predicting weather patterns, with a notable overall accuracy of 94.57%. These metrics affirm the classifier's suitability for this multi-class classification task.

4.3 Recommendations

The K-Means clustering model can be applied in real-world scenarios for weather-based resource allocation and planning. For instance, identifying weather patterns such as Hot_Dry or Cool_Humid can assist agricultural sectors in optimising irrigation schedules, crop selection, and pest management strategies. Additionally, urban planning authorities can use these insights for designing infrastructure that accommodates seasonal variations in weather conditions.

The Naive Bayes classifier, with its strong predictive performance, can be deployed in weather alert systems to classify and predict upcoming weather conditions based on real-time meteorological data. This can enhance disaster preparedness, such as issuing timely alerts for extreme weather events or optimising logistics and transportation systems under varying weather conditions. These applications demonstrate the practical relevance of our models in supporting decision-making across multiple domains.

5.0 Problems and Pitfalls

Throughout the course of analysis for this dataset, we encountered several challenges and limitation that worth highlight for future improvements:

Data Collection and Quality Issues:

- The dataset lacks certain meteorological variables, such as wind speed and solar radiation, which could have provided a more comprehensive analysis of weather patterns analysis. Including these variables would enable a more understanding of the atmosphere and their impact on temperature and humidity.
- The dataset is limited to specific locations within the City of Canning, which may not represent the entire city's microclimate variations.
- Weather data naturally contains outliers (extreme values) that we could not tell whether it was legitimate or just measurement errors. In our case, we chose to keep the supposed outliers as it seemed plausible.

Technical Limitations:

- Limited ability to handle extreme weather events due to their rare occurrence in the training data.
- Processing large weather datasets required careful memory management, as certain algorithms that we have tried were too computationally intensive and had to be discarded.

6.0 Conclusion

In conclusion, this project that we have carried out successfully demonstrated the application of machine learning techniques to analyze and classify weather patterns using meteorological data from the City of Canning. Through a two-stage modeling approach that integrated K-means clustering and Naive Bayes classification, we identified distinct weather patterns and developed a predictive model with strong accuracy and reliability. The clustering analysis provided meaningful insights into the natural groupings of weather conditions, while the classification model effectively translated these insights into actionable predictions.

Despite the various problems and limitations that we encountered, the results underscore the potential of data-driven approaches in enhancing weather analysis and decision-making processes across various sectors. Overall, we have learned a lot from attempting this project and we hope that it can highlight the transformative role of machine learning in uncovering patterns within complex datasets and driving informed decisions in the future.

References

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3. Priambodo, B., Jokonowo, B., Samidi, Ahmad, A., & Kadir, R. A. (2023). Predict traffic state based on PCA-KMeans clustering of neighbouring roads. *Lecture Notes in Computer Science*, 430–442. https://doi.org/10.1007/978-981-99-7339-2_36